

CADOGAN AGCM, AB, Canada

WMO: 710120

Lat: 52.3342N

Lon: 110.5100W

Elev: 2264

StdP: 13.53

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.5	-18.2	-29.6	1.1	-23.1	-23.9	1.6	-17.9	27.6	31.0	24.6	30.5	6.7	260	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	85.3	62.9	81.6	61.8	78.5	60.5	66.4	78.8	64.5	76.6	62.8	74.3	10.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	90.7	71.2	60.2	84.6	68.6	58.4	79.3	67.1	32.3	78.6	30.8	77.0	29.5	74.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	22.3	19.6	DB	-31.1	91.8	4.4	3.6	-34.3	94.4	-36.9	96.4	-39.4	98.4	-42.6	101.0	(4)
(5)			WB	-31.3	70.3	4.4	2.7	-34.5	72.2	-37.1	73.8	-39.6	75.3	-42.8	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	12.5	11.2	23.9	38.4	51.8	59.2	63.9	62.4	53.6	40.2	24.2	10.8		
	DBStd	22.47	15.51	14.56	14.01	9.48	7.96	5.89	5.01	5.83	8.42	9.14	12.81	14.45		
	HDD50	5872	1162	1087	810	361	74	6	0	1	57	324	775	1215		
	HDD65	10032	1627	1507	1275	798	411	188	81	119	352	770	1224	1680		
	CDD50	1425	0	0	0	12	131	283	431	384	164	19	1	0		
	CDD65	111	0	0	0	0	3	15	47	37	9	0	0	0		
	CDH74	1922	0	0	0	6	135	256	653	604	260	8	0	0		
	CDH80	480	0	0	0	0	18	49	175	154	84	0	0	0		
(14) Wind	WSAvg	9.5	10.1	9.6	10.0	10.9	9.9	9.1	8.4	7.9	8.7	9.7	9.7	9.5		
(15) Precipitation	PrecAvg	15.2	0.7	0.4	0.6	1.0	1.6	2.9	2.6	2.0	1.1	0.7	0.6	0.5		
	PrecMax	20.9	1.7	0.9	1.5	3.0	3.2	5.3	5.3	4.7	3.1	1.6	1.8	1.1		
	PrecMin	9.4	0.2	0.1	0.0	0.2	0.4	0.5	1.2	0.2	0.0	0.1	0.0	0.1		
	PrecStd	3.4	0.5	0.2	0.4	0.7	0.9	1.2	1.0	1.2	1.0	0.4	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	44.1	59.2	73.1	82.5	85.8	89.3	88.8	88.2	73.8	57.9	44.3		
		MCWB	37.4	37.7	45.7	50.3	56.5	61.7	65.7	63.7	60.4	54.9	44.8	37.4		
	2%	DB	40.5	39.0	51.8	66.8	77.9	80.0	84.9	84.0	81.4	64.6	49.5	39.2		
		MCWB	35.6	34.7	40.8	47.8	55.1	60.3	65.4	62.7	58.9	50.5	40.7	34.4		
	5%	DB	37.8	35.5	46.2	60.5	73.5	76.6	80.8	80.6	75.5	59.3	43.9	34.4		
		MCWB	33.6	31.9	38.8	44.4	53.7	59.4	63.7	62.5	57.1	47.4	37.3	30.5		
	10%	DB	34.1	32.1	40.9	55.2	69.0	73.1	77.2	77.0	69.9	54.5	39.7	30.1		
		MCWB	31.1	29.6	35.5	42.2	51.5	57.8	62.2	61.2	54.6	44.7	34.8	27.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	39.0	46.6	51.3	59.1	65.7	70.5	68.3	62.6	56.4	46.3	38.4		
		MCDB	43.7	43.8	59.4	71.1	77.0	78.8	82.9	80.9	80.3	71.7	55.9	43.1		
	2%	WB	36.1	34.9	42.1	48.1	56.6	63.1	67.2	65.9	60.1	52.2	41.7	34.3		
		MCDB	40.2	38.5	50.2	65.4	73.4	76.1	79.0	79.3	78.2	61.5	48.7	39.0		
	5%	WB	33.8	32.3	39.0	45.3	54.7	61.0	65.3	63.9	57.8	48.6	38.1	30.6		
		MCDB	37.5	35.3	45.5	58.2	70.3	72.5	76.7	76.3	72.7	57.8	42.7	33.9		
	10%	WB	31.0	29.4	35.9	42.9	52.7	59.1	63.6	62.2	55.7	45.6	35.3	27.6		
		MCDB	34.2	32.2	40.8	54.1	66.9	70.0	74.5	73.6	69.1	53.8	39.1	30.2		
(35) Mean Daily Temperature Range	MDBR		17.3	17.8	18.0	21.3	24.5	22.0	23.2	23.6	24.2	20.3	16.2	15.6		
	5% DB	MCDBR	18.0	20.1	24.1	31.4	31.1	27.9	29.6	31.0	34.2	28.9	21.5	19.1		
		MCWBR	14.1	16.1	15.1	15.5	13.5	11.8	13.0	12.9	15.0	15.8	14.4	15.1		
	5% WB	MCDBR	18.4	19.9	23.2	29.0	28.2	24.0	26.1	26.5	31.4	27.2	19.7	18.6		
		MCWBR	14.9	16.2	15.1	14.9	13.0	11.3	12.7	12.0	14.4	15.2	14.0	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.293	0.290	0.337	0.362	0.373	0.365	0.378	0.321	0.284	0.259	0.241		
	taud		2.347	2.256	2.436	2.343	2.322	2.331	2.394	2.327	2.458	2.539	2.461	2.355		
	Ebn at Noon		245	260	290	285	282	279	279	268	275	267	242	229		
	Edh at Noon		18	27	28	35	38	38	35	36	28	21	16	15		
(44) All-Sky Solar Radiation	RadAvg		354	688	1146	1513	1837	1898	2014	1637	1159	695	357	265		
	RadStd		31	53	86	141	124	129	97	132	118	92	35	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page